Fundamentals of Data Science – Small Project

Name : Jency Francis

Student ID : 23082639

Introduction

The analysis was conducted on a dataset titled “airline3.csv”, which displays the daily performance metrics of an airline during 2021,2022 and 2023. This data is formatted as a comma-separated values (CSV) file and contains three columns: the date of the flight (in DD-MM-YYYY format), the number of passengers carried on that specific day (in thousands), and the daily average ticket price (in Euros). The type of both the ticket Price and Number of passengers are ‘float’ and Date is an ‘object’.

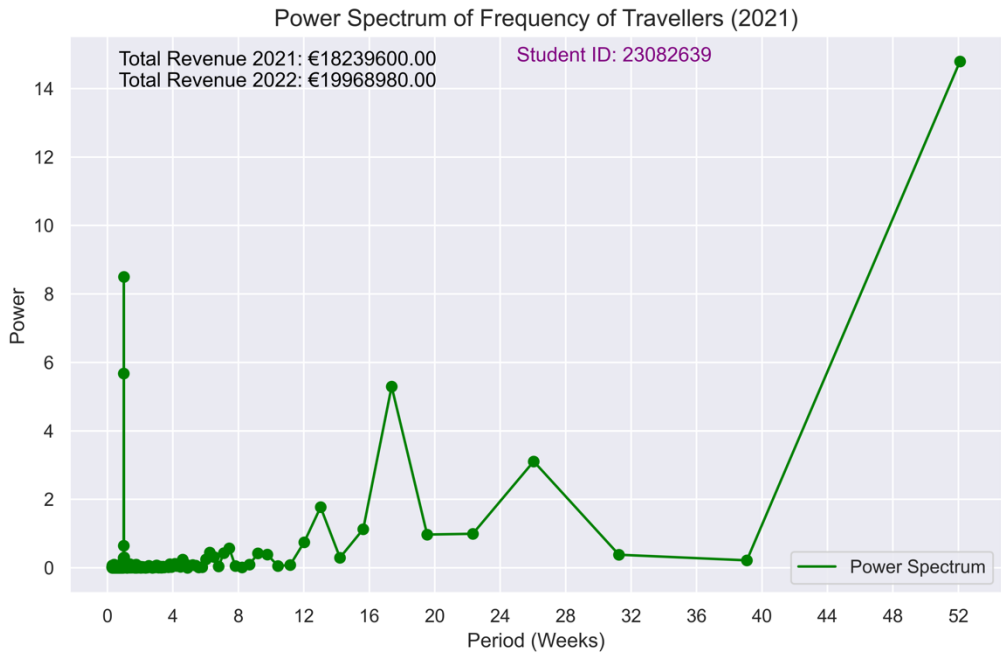
Figure 1:

A Fourier transform is a powerful mathematical approach that converts a signal from the time domain to the frequency domain. Visualizing the Fourier transform with a bar plot combines raw data with the Fourier-fitted model and helps us to understand how accurately the Fourier series displays the periodic behavior of the data. The Fourier Fit Line is a curve that depicts the fitted Fourier series, precisely indicating the periodic trends and patterns present in the dataset. In this bar plot, the x-axis represents the ‘Month’, while the y-axis captures ‘Total Passengers (in Thousands)’. The graph has been created for three years: 2021, 2022 and 2023, the Fourier Fit is done for 2021 only. Observations reveal a peak in July, followed by August, due to the summer season. Subsequent peaks in April and December show increase, with a decline afterwards, and then a rise again towards the year's end, owing to festive holidays.

Figure 2:

Power Spectrum plays a crucial role in identifying the frequencies that are key to shaping the overall signal, making it an asset for revealing most influential periodic patterns. We can effectively highlight the frequencies that notably influence the signal basis its dominant frequencies. It allows us to differentiate between long-term trends (low frequencies) and

short-term fluctuations (high frequencies) based on its signal structure. This visualization allows for easy interpretation of the signal's frequency content, representing the signal as a sum of sine and cosine waves.In this power spectrum of Frequency of Travellers for the year 2021, the 'Period(Weeks)' has been captured on X-axis and Power is on the Y-axis. As per the plot, it can be observed that, during the beginning of the year (in January) the number of passengers is on the higher side. However, proceeds to become flat almost till 11th week. Also, it starts to increase gradually from 39th week towards the end of December and it touches it peak value at 52nd week which is the dominant period owing to festivities. Total Revenue 2021(X = €18239600.00) and Total Revenue 2022 (Y = €19968980.00) also have been captured in the plot.



Important Formula:

1) X = Total Revenue for 2021 = $\sum(total\ number\ for\ 2021) * (Total\ Price\ for\ 2021) * 1000$

2) Y = Total Revenue for 2022 = $\sum(\ total\ number\ for\ 2022) * (Total\ Price\ for\ 2022) * 1000$

3) Power = $ccoef^2$ + $scoef^2$

4) Fourier Series = f (x) = $\sum_{k=0}^{\infty}\left[A_K \cos\left(\frac{2\pi k}{L}x\right) + B_k \sin\left(\frac{2\pi k}{L}x\right)\right]$

Conclusion:

To conclude, it can be said that we observe the highest peak in July and August months due to summer season and in April and December basis the Fourier Analysis and Fit. The power spectrum displays the prediction of passenger travel, following a trend on a recurring pattern every year with December being the highest followed by January being the second highest.